HGCNet: Hierarchical Global–Local Collaborative Network for Hyperspectral Image Classification

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***Abstract*—Global contextual modeling and local detail preservation are both essential for hyperspectral image (HSI) classification, as they jointly determine the discriminability of spectral–spatial representations. Effective global modeling captures long-range contextual dependencies and overall semantic structures, whereas careful use of local information is critical for preserving fine-grained details, such as edges, textures, and subtle spatial variations. Although recent global modeling techniques, such as Mamba, can efficiently capture long-range dependencies with linear computational complexity, they often overlook or weaken local structural details, leading to incomplete feature representations. To address this limitation, this article proposes a hierarchical global–local collaborative network (HGCNet), a HGCNet for HSI classification. Specifically, HGCNet is built on a hierarchical spectral–spatial collaborative framework, in which spatial and spectral representations are progressively learned and jointly optimized within a unified architecture. To enhance spatial feature learning, a hybrid spatial recalibration module (HSRM) is designed to jointly capture global contextual dependencies and local structural details. Meanwhile, a dual-path spectral recalibration module (DSRM) is developed to collaboratively model global spectral correlations and local band-wise variations, thereby improving spectral discriminability. Furthermore, a bidirectional spectral–spatial interaction module (BSSIM) is introduced to promote interaction between spatial and spectral features, thereby strengthening their complementary representation capability. By jointly exploiting global and local information and enhancing spectral–spatial interaction, HGCNet learns more discriminative and robust feature representations for HSI classification. Extensive experiments on four benchmark datasets demonstrate that the proposed method achieves superior classification performance over existing advanced methods. The code will be available online at https://github.com/YichuXu/HGCNet**

***Index Terms*—Bidirectional interaction fusion, global–local collaborative modeling, hyperspectral image (HSI) classification, Mamba, spectral–spatial learning.**

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I. Introduction

HYPERSPECTRAL imaging technology combines the advantages of narrow-band imaging and imaging spectroscopy, enabling the acquisition of dozens to hundreds of contiguous spectral channels across the visible, near-infrared, and short-wave infrared regions [1], [2], [3]. Each pixel contains not only spatial location information but also an almost continuous spectral signature, forming a distinctive 3-D spectral–spatial data cube [4], [5]. Compared with conventional optical RGB imagery, hyperspectral data provide substantially stronger capability for fine-grained recognition and interpretation [6], [7]. Owing to their high spectral resolution and dense band structure, hyperspectral images (HSIs) can capture the unique spectral fingerprints of land-cover materials, enabling effective discrimination of subtle categories and material components that are difficult to identify in traditional panchromatic or multispectral imagery. Leveraging their distinctive spectral characteristics, HSIs support a broad range of remote sensing tasks, including classification [8], [9], target detection [10], [11], [12], change detection [13], [14], and image restoration [15], [16], [17].

Among these application scenarios, HSI classification is one of the most fundamental and essential tasks, as it serves as a prerequisite for fine-grained scene interpretation and efficient utilization of hyperspectral information [18], [19]. The objective of this task is to assign accurate semantic labels to each pixel in an image through the joint analysis of spectral characteristics and spatial contextual information, ultimately producing a high-precision land-cover map.

With the rapid development of deep learning, HSI classification has gradually evolved from handcrafted feature engineering methods, such as tensor singular spectrum analysis [20], support vector machines [21], and random forests [22], to fully automatic neural network-based approaches. Beyond conventional supervised classification, cross-domain HSI classification has attracted growing attention for its role in improving robustness to domain shifts caused by sensor differences, imaging conditions, and scene variations. Existing studies have examined bi-directional domain adaptation to reduce source-target discrepancies [23], single-source domain expansion and language-aware domain generalization to improve cross-scene robustness [24], [25], and spectral foundation models for transferable spectral representation

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learning [26]. These efforts indicate that robust and discriminative spectral–spatial representation learning remains essential for HSI classification in diverse scenarios. Early convolutional neural networks (CNNs) [27] demonstrated strong capability in local feature extraction; however, their limited receptive fields restricted the modeling of global contextual information. Subsequently, Transformer-based models [28], [29] introduced long-range dependency modeling, thereby alleviating the inherent limitation of CNNs in contextual perception. Nevertheless, their reliance on self-attention mechanisms [30], [31] incurs quadratic computational complexity, resulting in low efficiency when jointly modeling spatial and spectral contexts. Recently, diffusion-based methods [32], [33] and foundation model-based approaches [34] have also been explored for HSI classification, showing promise in generative spectral–spatial modeling and transferable representation learning, respectively, but often with higher computational or pretraining costs. In contrast, the emergence of Mamba [35], [36], [37] has attracted increasing attention in HSI classification, as its architecture enables contextual modeling while maintaining linear computational complexity. However, Mamba is mainly effective in modeling sequence-level dependencies, while its ability to preserve local spectral–spatial structures remains limited.

Recent global modeling methods, including Mamba, can capture long-range dependencies with linear computational complexity. However, they may weaken local structural details and produce incomplete feature representations. This limitation is closely related to the way spatial feature maps or spectral–spatial cubes are converted into scanned sequences, where contextual information is mainly propagated through global state updates. This process helps model long-range relationships and maintain semantic consistency, but it does not directly focus on local neighborhood structures. Consequently, fine-grained local variations may be lost during global feature aggregation. In HSI classification, such local cues are important because different land-cover classes can have very similar spectral responses, and their separation often depends on subtle spatial patterns and local spectral differences. Therefore, effective HSI representation learning should combine global context with local fine-grained information, rather than relying only on global sequence modeling. To address this limitation, this article proposes a hierarchical global–local collaborative network (HGCNet) for HSI classification. HGCNet is built on a hierarchical spectral–spatial collaborative framework, in which spatial and spectral representations are progressively learned and jointly optimized. Specifically, a hybrid spatial recalibration module (HSRM) is designed in the spatial branch, where a visual state space model is employed to capture global spatial dependencies, while convolutional operations are introduced to extract locally enhanced spatial features. A gating mechanism is further adopted to adaptively integrate global and local spatial representations, enabling the network to preserve both global spatial semantics and fine-grained texture details. Furthermore, a dual-path spectral recalibration module (DSRM) is developed in the spectral branch to enhance spectral representation through collaborative global–local modeling. Specifically, it combines global dependency modeling across bands with local spectral relation learning among adjacent bands, and adaptively fuses the resulting spectral responses to recalibrate channel-wise features. In this way, the module effectively captures both interband correlations and fine-grained spectral variations, thereby improving spectral discriminability and sensitivity to subtle spectral differences. A bidirectional spectral–spatial interaction module (BSSIM) is further introduced to perform mutual refinement between the spatial and spectral branches. By allowing spectral responses to modulate spatial feature aggregation and spatial cues to guide spectral feature enhancement, the module establishes bidirectional spectral–spatial fusion and promotes more coordinated cross-domain representation learning. The main contributions of this work can be summarized as follows.

1) This article proposes a novel HGCNet for HSI classification. HGCNet is built on a hierarchical spectral–spatial collaborative framework, in which spectral and spatial representations are progressively learned and jointly optimized within a unified architecture. By integrating global–local modeling and interactive learning between spectral and spatial features, the proposed network learns more discriminative spectral–spatial representations.
2) This article designs an HSRM to improve spatial representation learning through global–local collaborative modeling. By combining global spatial dependency modeling with local structural detail extraction, HSRM enables the network to capture long-range contextual information and fine-grained texture patterns.
3) This article develops a DSRM to enhance spectral feature learning from global and local perspectives. DSRM jointly models global spectral correlations and local band-wise variations to adaptively recalibrate spectral responses, thereby improving the representation of spectral dependencies and subtle interband differences.
4) A BSSIM is introduced to enhance the fusion of spatial and spectral features. By jointly leveraging channel-wise spectral guidance and spatial-wise spatial guidance, BSSIM strengthens the complementary interaction between spatial structural information and spectral discriminative information, leading to more effective spectral–spatial representation learning.

The rest of this article is organized as follows. Section II surveys related studies on HSI classification. Section III details the proposed HGCNet. Experimental results and corresponding analyses are presented in Section IV. Section V concludes this article.

II. Related Work

In this section, we briefly review the CNN-based, Transformer-based, and Mamba-based methods in the HSI classification community.

A. CNN-Based HSI Classification

CNNs [38] are among the earliest deep models used for HSI classification, replacing handcrafted features with hierarchical representations learned through stacked convolutions. Hu et al.

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[39] present a 1-D-CNN as an early attempt to model local spectral features. To better exploit spectral–spatial information, subsequent studies extend CNNs to 2-D-CNNs [27] and 3-D-CNNs [40]. On this basis, Xu et al. [41] proposed SSUN, which enhances spectral–spatial representation through multiscale feature learning. Beyond patch-level methods, SSFCN [42] is introduced as the first image-level framework for HSI classification, employing a dual-branch fully convolutional network to model spectral and spatial features separately. To further improve classification efficiency, Zheng et al. [43] proposed the fast patch-free global learning framework (FPGA), which strengthens discriminative spectral features through an encoder equipped with spectral attention and combines it with a lightweight decoder to balance efficiency and performance. Later studies further improve FCN-based models by incorporating multiscale designs and attention mechanisms to strengthen feature extraction and fusion [44]. Nevertheless, due to the locality of convolution operations, CNN-based methods still have limited ability to capture long-range contextual dependencies, which constrains the learning of sufficiently comprehensive and discriminative representations.

B. Transformer-Based HSI Classification

In recent years, Transformer [45] architectures have been widely adopted for remote sensing tasks because they can effectively model long-range dependencies. Hong et al. [28] proposed SpectralFormer, which reformulates HSI classification as a sequence modeling problem and learns spectral information from the sequential relations among adjacent bands. However, this method mainly focuses on spectral modeling and makes limited use of spatial information, which restricts its ability to characterize complex HSI scenes. To overcome this limitation, subsequent studies develop a series of spectral–spatial transformer models that jointly model spectral and spatial features. These methods improve feature representation through multiscale spectral–spatial learning [46], [47], hierarchical feature modeling [48], and more effective use of contextual information [29]. Some studies further combine convolution with Transformer architectures to balance local detail modeling and global context modeling. Zhang et al. [49] proposed CTMixer, which enhances the joint modeling of local and global information through collaborative convolution and Transformer modules. In contrast to such hybrid designs, Roy et al. [50] proposed MorphFormer, which incorporates morphological structural information into the attention mechanism to improve feature representation. For lightweight modeling, Zhao et al. [51] proposed GSC-ViT, which integrates convolution and self-attention in a compact manner, reducing model complexity while retaining both local detail modeling and global contextual representation.

Despite these advances, most Transformer-based methods still rely on local image patches because the quadratic complexity of self-attention makes full-image modeling costly in computation and memory. However, this localized processing is not fully consistent with the strength of Transformers in modeling global dependencies and thus limits the use of global contextual information. To address this issue, recent studies explore Transformer architectures with whole-image input. Zou et al. [52] proposed LESSFormer, which enhances local dependencies while preserving long-range context, thereby improving feature representation for HSI classification. Wang et al. [53] further proposed DCN-T, which incorporates external prior information and region-level contextual modeling to improve scene representation and prediction accuracy.

C. Mamba-Based HSI Classification

Very recently, Mamba [54], [55] has attracted increasing attention in HSI classification because it can model long-range dependencies with linear computational complexity. Existing studies mainly explore this advantage from the perspectives of spectral–spatial joint modeling, selective scanning design, and multiscale feature extraction. Yao et al. [56] proposed Spectral-Mamba, which combines dynamic spectral–spatial fusion with selective state-space modeling to alleviate spectral variability and class confusion while enabling efficient pixel-level and patch-level classification. Huang et al. [57] further proposed SS-Mamba, which represents hyperspectral data as spatial and spectral sequences and enhances long-range representation through stacked spectral–spatial Mamba blocks and an interaction enhancement mechanism. Subsequently, He et al. [58] developed 3DSS-Mamba, which represents hyperspectral cubes as 3-D spectral–spatial tokens and explicitly models global long-range spectral–spatial dependencies while preserving linear complexity. To address high-dimensional redundancy and the demand for multiscale modeling, He et al. [59] further proposed IGroupSS-Mamba, which improves the efficiency of global spectral–spatial information utilization through grouped modeling and multiscale feature extraction.

Beyond these methods, several studies further extend the application of Mamba to hyperspectral classification through improvements in scanning strategy, structural modeling, and feature fusion. Liu et al. [60] proposed HyperMamba, which strengthens spectral–spatial interaction through adaptive spatial and spectral scanning. Yang et al. [61] proposed GraphMamba, which introduces graph structure learning into the visual Mamba framework to better capture dynamic spatial structural context and enhance spectral–spatial joint representation. Wang et al. [62] proposed S2Mamba, which performs contextual modeling in both the spatial and spectral dimensions and uses a hybrid gating mechanism to adaptively fuse the two types of features, thereby achieving efficient and accurate classification with fewer parameters.

As research progresses, some studies move beyond the local patch input paradigm and begin to explore whole-image Mamba frameworks in order to better exploit the advantage of Mamba in global context modeling. Li et al. [63] proposed MambaHSI, the first whole-image Mamba framework for hyperspectral classification, which learns joint discriminative spectral–spatial representations with global contextual consistency through dual-path state-space modeling of spatial and spectral information. On this basis, Pan et al. [64] proposed MambaLG, which addresses the loss of local details under whole-image input by constructing a local-to-global dual-branch framework, thereby balancing local structure

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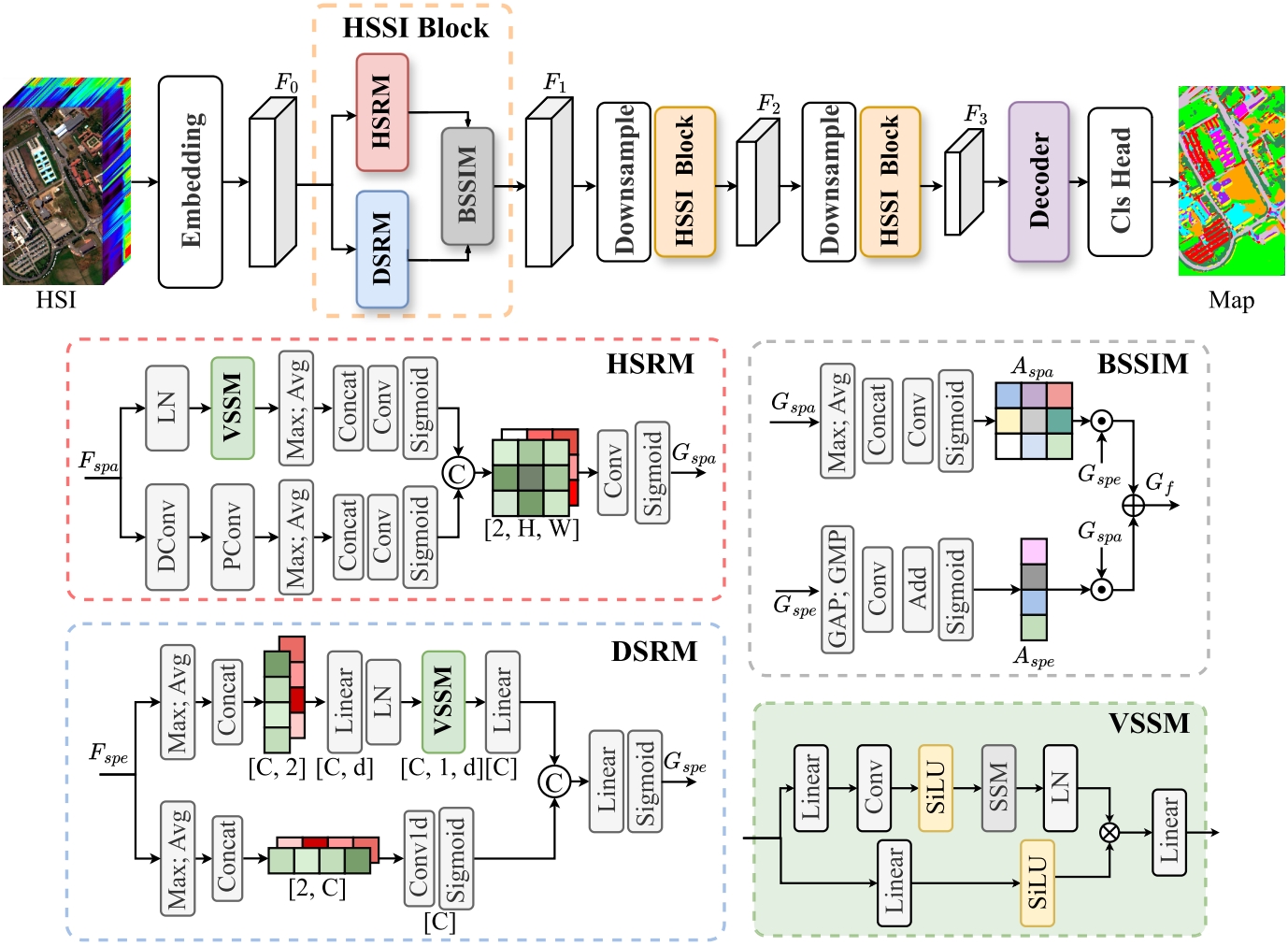


Fig. 1. Overall architecture of the proposed HGCNet. After feature embedding, stacked HSSI Blocks are used to hierarchically extract spectral–spatial representations, where each block consists of an HSRM, a DSRM, and a BSSIM, with VSSMs embedded in HSRM and DSRM. Finally, a lightweight decoder and a classification head generate the pixel-wise classification map.

preservation and global semantic representation. Xu et al. [65] further proposed PHDMamba, which progressively captures long-range spectral–spatial contextual dependencies and gradually fuses spectral and spatial information through adaptive feature interaction, thus further enhancing representation capability.

III. Method

A. Overview

Given an input HSI cube $\mathbf{X} \in \mathbb{R}^{C_{\text{in}} \times H \times W}$, where C_{in}, H, and W denote the number of spectral bands, height, and width, respectively, the proposed network aims to learn discriminative spectral–spatial representations for dense HSI classification. As illustrated in Fig. 1, the overall architecture follows a hierarchical encoder–decoder pipeline and mainly consists of three components: 1) a shallow embedding layer for channel projection; 2) a multistage spectral–spatial representation extractor built upon hierarchical spectral–spatial interaction modules (HSSI Blocks); and 3) a top-down feature aggregation decoder followed by a classification head.

Specifically, the input HSI is first fed into an embedding layer implemented by a 1×1 convolution, followed by group normalization and a SiLU activation, to project the original spectral dimension into a compact latent feature space

$$\mathbf{F}_0 = \mathcal{E}(\mathbf{X}) \tag{1}$$

where $\mathcal{E}(\cdot)$ denotes the embedding operator, and $\mathbf{F}_0 \in \mathbb{R}^{C \times H \times W}$ is the embedded feature map with C channels.

Based on the embedded features, the network stacks three hierarchical HSSI Blocks to progressively extract multiscale representations. The first HSSI Block operates at the original spatial resolution to produce $\mathbf{F}_1$, while the subsequent two stages first apply downsampling and then perform feature extraction, yielding $\mathbf{F}_2$ and $\mathbf{F}_3$, respectively. Formally, the encoding process can be written as

$$\mathbf{F}_1 = \mathcal{B}_1(\mathbf{F}_0) \tag{2}$$

$$\mathbf{F}_2 = \mathcal{B}_2(\mathcal{D}_1(\mathbf{F}_1)) \tag{3}$$

$$\mathbf{F}_3 = \mathcal{B}_3(\mathcal{D}_2(\mathbf{F}_2)) \tag{4}$$

where $\mathcal{B}_i(\cdot)$ denotes the ith HSSI Block and $\mathcal{D}_i(\cdot)$ denotes the downsampling operator implemented by average pooling. Accordingly, the feature maps at different stages are of sizes $\mathbf{F}_1 \in \mathbb{R}^{C \times H \times W}$, $\mathbf{F}_2 \in \mathbb{R}^{C \times H/2 \times W/2}$, and $\mathbf{F}_3 \in \mathbb{R}^{C \times H/4 \times W/4}$.

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Each HSSI Block consists of an HSRM, a DSRM, and a BSSIM. The HSRM and DSRM are used to encode spatial and spectral information, respectively, while the BSSIM is introduced to integrate the two types of representations. In addition, vision state space modules (VSSMs) are incorporated into the HSSI Block to enhance long-range context modeling with linear computational complexity.

After obtaining multilevel encoder features $\{\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3\}$, a lightweight top-down decoder is employed to progressively fuse deep semantic features with shallow detailed features. Starting from the deepest representation $\mathbf{F}_3$, the decoder successively upsamples and aggregates features from the preceding stages

$$\hat{\mathbf{F}}_3 = \mathcal{P}_1\left(\mathbf{F}_3\right) \tag{5}$$

$$\hat{\mathbf{F}}_2 = \mathcal{P}_2\left(\hat{\mathbf{F}}_3, \mathbf{F}_2\right) \tag{6}$$

$$\hat{\mathbf{F}}_1 = \mathcal{P}_3\left(\hat{\mathbf{F}}_2, \mathbf{F}_1\right) \tag{7}$$

where $\mathcal{P}_i(\cdot)$ denotes the feature fusion block. This top-down aggregation strategy enables the final representation to combine high-level semantic information with fine-grained spatial details.

Finally, the fused feature is resized to the original input resolution and fed into a classification head composed of two 1×1 convolutions, normalization, and nonlinear activation to generate the pixel-wise prediction map

$$\mathbf{Y} = \mathcal{H}\left(\hat{\mathbf{F}}_1\right) \tag{8}$$

where $\mathcal{H}(\cdot)$ denotes the classification head, $\mathbf{Y} \in \mathbb{R}^{N_c \times H \times W}$, and N_c is the number of categories.

B. Hybrid Spatial Recalibration Module

To capture both global contextual information and local spatial details for comprehensive spatial modeling, we introduce an HSRM. The module is composed of two complementary branches, namely a global spatial perception branch and a local spatial refinement branch. The former models long-range spatial dependencies by leveraging the VSSM, while the latter focuses on fine-grained neighborhood structures through convolutional operations. The outputs of these two branches are subsequently transformed into spatial attention maps and fused to generate a unified spatial recalibration factor.

Given the input spatial feature $\mathbf{F}_{\text{spa}} \in \mathbb{R}^{C \times H \times W}$, the global branch first employs a VSSM-based operator to model long-range spatial interactions

$$\mathbf{F}_g = \text{VSSM}\left(\mathbf{F}_{\text{spa}}\right) \tag{9}$$

where VSSM$(\cdot)$ denotes the VSSM. To summarize the informative responses of $\mathbf{F}_g$ along the channel dimension, average-pooling and max-pooling are first performed, and the resulting descriptors are concatenated and fed into a convolutional projection followed by a sigmoid activation to produce a global spatial attention map:

$$\mathbf{A}_g = \sigma\left(\phi_g\left(\left[\text{Avg}\left(\mathbf{F}_g\right), \text{Max}\left(\mathbf{F}_g\right)\right]\right)\right) \tag{10}$$

where Avg$(\cdot)$ and Max$(\cdot)$ denote channel-wise average-pooling and max-pooling, respectively, $[\cdot,\cdot]$ denotes concatenation, $\phi_g(\cdot)$ represents the convolutional transformation, and $\sigma(\cdot)$ is the sigmoid activation.

Meanwhile, the local branch applies depthwise and pointwise convolutions to extract fine-grained neighborhood patterns

$$\mathbf{F}_l = \mathcal{L}\left(\mathbf{F}_{\text{spa}}\right) \tag{11}$$

where $\mathcal{L}(\cdot)$ denotes the local spatial refinement operator. In the same manner, the local feature is converted into a local spatial attention map

$$\mathbf{A}_l = \sigma\left(\phi_l\left(\left[\text{Avg}\left(\mathbf{F}_l\right), \text{Max}\left(\mathbf{F}_l\right)\right]\right)\right). \tag{12}$$

The two attention maps are then fused to form a unified spatial recalibration map

$$\mathbf{G}_{\text{spa}} = \sigma\left(\phi_f\left(\left[\mathbf{A}_g, \mathbf{A}_l\right]\right)\right) \tag{13}$$

where $\phi_f(\cdot)$ denotes the fusion convolution. Finally, the fused attention map is applied to recalibrate the input feature in an element-wise manner

$$\hat{\mathbf{F}}_{\text{spa}} = \mathbf{F}_{\text{spa}} \odot \mathbf{G}_{\text{spa}} \tag{14}$$

where $\odot$ denotes element-wise multiplication.

C. Dual-Path Spectral Recalibration Module

Different from spatial modeling, spectral representation learning requires the network to characterize both global inter-band dependencies and compact statistical cues. To this end, we devise a DSRM. This module contains two parallel paths. One path constructs compact spectral tokens and employs the VSSM to model long-range spectral dependencies, while the other directly exploits pooled spectral statistics to encode complementary channel-wise priors. Their outputs are subsequently fused to generate a spectral recalibration vector.

Given the input spectral feature $\mathbf{F}_{\text{spe}} \in \mathbb{R}^{C \times H \times W}$, spatial average-pooling and max-pooling are first conducted to summarize each spectral channel

$$\mathbf{f}_{\text{avg}} = \text{Avg}_s\left(\mathbf{F}_{\text{spe}}\right), \quad \mathbf{f}_{\text{max}} = \text{Max}_s\left(\mathbf{F}_{\text{spe}}\right) \tag{15}$$

where $\mathbf{f}_{\text{avg}}, \mathbf{f}_{\text{max}} \in \mathbb{R}^C$, and $\text{Avg}_s(\cdot)$ and $\text{Max}_s(\cdot)$ denote spatial average-pooling and spatial max-pooling, respectively.

In the first path, the two spectral descriptors are stacked to form a compact token sequence

$$\mathbf{T} = \left[\mathbf{f}_{\text{avg}}, \mathbf{f}_{\text{max}}\right] \in \mathbb{R}^{C \times 2}. \tag{16}$$

A linear projection is then used to map the 2-D statistics of each channel into a higher dimensional latent space

$$\mathbf{Z} = \psi\left(\mathbf{T}\right) \in \mathbb{R}^{C \times d} \tag{17}$$

where $\psi(\cdot)$ denotes the linear projection and d is the token dimension. After layer normalization, the projected tokens are fed into a VSSM to capture long-range spectral dependencies

$$\mathbf{Z}_g = \text{VSSM}\left(\text{LN}\left(\mathbf{Z}\right)\right). \tag{18}$$

The resulting contextualized tokens are further projected into a channel-wise global spectral gate

$$\mathbf{g}_1 = \sigma\left(\psi_g\left(\mathbf{Z}_g\right)\right) \tag{19}$$

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where $\psi_g(\cdot)$ denotes a linear transformation and $\mathbf{g}_1 \in \mathbb{R}^C$.

In the second path, the same pooled descriptors are directly aggregated and passed through a lightweight convolutional operator to model statistical interactions among spectral channels

$$\mathbf{S} = \left[\mathbf{f}_{\text{avg}}, \mathbf{f}_{\text{max}}\right] \tag{20}$$

$$\mathbf{g}_2 = \sigma\left(\phi_s\left(\mathbf{S}\right)\right) \tag{21}$$

where $\phi_s(\cdot)$ denotes the convolution-based statistical modeling operator and $\mathbf{g}_2 \in \mathbb{R}^C$ is the statistical spectral gate.

After that, the two spectral gates are concatenated and fused to generate the final spectral recalibration vector

$$\mathbf{G}_{\text{spe}} = \sigma\left(\psi_f\left(\left[\mathbf{g}_1, \mathbf{g}_2\right]\right)\right) \tag{22}$$

where $\psi_f(\cdot)$ denotes the fusion projection and $\mathbf{G}_{\text{spe}} \in \mathbb{R}^C$. Finally, the obtained spectral gate is used to recalibrate the input feature in a channel-wise manner

$$\hat{\mathbf{F}}_{\text{spe}} = \mathbf{F}_{\text{spe}} \odot \mathbf{G}_{\text{spe}} \tag{23}$$

where $\mathbf{G}_{\text{spe}}$ is broadcast along the spatial dimensions and $\odot$ denotes element-wise multiplication.

D. Bidirectional Spectral–Spatial Interaction Module

To effectively integrate the complementary information encoded by the spatial and spectral branches, we design a BSSIM. The core idea of BSSIM is to perform bidirectional guidance between the two branches, such that the spectral representation can be used to recalibrate the spatial feature, while the spatial representation can also be exploited to refine the spectral feature. In this way, the module promotes more sufficient spectral–spatial interaction and yields a unified feature representation with enhanced discriminability.

Given the spatial feature $\mathbf{X}_{\text{spa}} \in \mathbb{R}^{C\times H\times W}$ and the spectral feature $\mathbf{X}_{\text{spe}} \in \mathbb{R}^{C\times H\times W}$, BSSIM first extracts a channel attention vector from the spectral branch to guide the spatial feature enhancement. Specifically, global average-pooling and global max-pooling are first applied to $\mathbf{X}_{\text{spe}}$

$$\mathbf{z}_{\text{avg}} = \text{GAP}\left(\mathbf{X}_{\text{spe}}\right), \quad \mathbf{z}_{\text{max}} = \text{GMP}\left(\mathbf{X}_{\text{spe}}\right) \tag{24}$$

where $\mathbf{z}_{\text{avg}}, \mathbf{z}_{\text{max}} \in \mathbb{R}^{C\times 1\times 1}$. These two descriptors are then fed into a shared lightweight channel transformation to generate the spectral-guided channel attention

$$\mathbf{A}_{\text{spe}} = \sigma\left(\phi_c\left(\mathbf{z}_{\text{avg}}\right) + \phi_c\left(\mathbf{z}_{\text{max}}\right)\right) \tag{25}$$

where $\phi_c(\cdot)$ denotes the channel projection implemented by two 1×1 convolutions with a nonlinear activation in between, and $\sigma(\cdot)$ is the sigmoid function. The obtained channel attention is then used to recalibrate the spatial feature

$$\tilde{\mathbf{X}}_{\text{spa}} = \mathbf{X}_{\text{spa}} \odot \mathbf{A}_{\text{spe}} \tag{26}$$

where $\odot$ denotes element-wise multiplication.

Meanwhile, BSSIM derives a spatial attention map from the spatial branch to guide the refinement of spectral features. Specifically, channel-wise average-pooling and max-pooling are performed on $\mathbf{X}_{\text{spa}}$

$$\mathbf{m}_{\text{avg}} = \text{Avg}_c\left(\mathbf{X}_{\text{spa}}\right), \quad \mathbf{m}_{\text{max}} = \text{Max}_c\left(\mathbf{X}_{\text{spa}}\right) \tag{27}$$

where $\mathbf{m}_{\text{avg}}, \mathbf{m}_{\text{max}} \in \mathbb{R}^{1\times H\times W}$. After concatenation, they are passed through a spatial convolution followed by a sigmoid activation to produce the spatial-guided attention map:

$$\mathbf{A}_{\text{spa}} = \sigma\left(\phi_s\left(\left[\mathbf{m}_{\text{avg}}, \mathbf{m}_{\text{max}}\right]\right)\right) \tag{28}$$

where $\phi_s(\cdot)$ denotes a 7×7 convolution and $[\cdot,\cdot]$ denotes channel concatenation. The resulting spatial attention is then used to enhance the spectral feature

$$\tilde{\mathbf{X}}_{\text{spe}} = \mathbf{X}_{\text{spe}} \odot \mathbf{A}_{\text{spa}}. \tag{29}$$

After bidirectional recalibration, the enhanced spatial and spectral features are aggregated by element-wise summation

$$\mathbf{X}_f = \tilde{\mathbf{X}}_{\text{spa}} + \tilde{\mathbf{X}}_{\text{spe}}. \tag{30}$$

Finally, a projection layer composed of a 1×1 convolution, group normalization, and GELU activation is employed to further align and refine the fused representation

$$\mathbf{Y} = \phi_p\left(\mathbf{X}_f\right) \tag{31}$$

where $\phi_p(\cdot)$ denotes the output projection operator.

IV. Experiment

A. Datasets

1) *Pavia University Dataset:* The dataset was acquired in 2001 using the Reflective Optics System Imaging Spectrometer (ROSIS) over the University of Pavia, located in northern Italy. It comprises 103 spectral bands with a spatial dimension of 610×340 pixels and provides annotations for nine distinct land-cover categories;
2) *Houston Dataset:* The University of Houston dataset, acquired in 2012, captures both the campus and surrounding urban areas. It comprises 144 spectral bands across the 400–1000-nm range, with a spatial resolution of 2.5 m. The imagery spans 349×1905 pixels and is labeled with 15 semantic categories [66];
3) *Whu-HongHu Dataset:* This dataset was collected using a UAV platform in HongHu City in 2017 [67]. It comprises 270 spectral bands ranging from 400 to 1000 nm, with a high spatial resolution of approximately 0.04 m. The dimensions of the imagery are 940×475 pixels, which includes 17 typical crop types;
4) *Whu-HanChuan Dataset:* The dataset, first introduced by Zhong et al. [67], was acquired using UAV-based hyperspectral imaging. It provides high spatial fidelity with centimeter-level resolution and comprises 274 spectral bands spanning the 400–1000-nm wavelength range. The spatial dimensions of the captured scene are 1217×303 pixels.

B. Experimental Settings

All experiments were conducted using the PyTorch framework on an NVIDIA GeForce RTX 3090 GPU. The training set is randomly selected according to the configurations in Tables I–IV. The training process employed the Adam optimizer with a learning rate of 1×10^{-3} and was conducted for 200 epochs across all datasets. Classification performance

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TABLE I
TRAINING AND TESTING SETTINGS OF THE PAVIA UNIVERSITY DATASET

Class ID	Class name	Color	Training	Testing	Total
1	Asphalt		15	6616	6631
2	Meadows		15	18634	18649
3	Gravel		15	2084	2099
4	Trees		15	3049	3064
5	Metal sheets		15	1330	1345
6	Bare soil		15	5014	5029
7	Bitumen		15	1315	1330
8	Bricks		15	3667	3682
9	Shadows		15	932	947
Total			135	41641	42776

TABLE II
CATEGORY AND SAMPLE SETTINGS OF THE HOUSTON DATASET

Class ID	Class name	Color	Training	Testing	Total
1	Grass healthy		15	1236	1251
2	Grass stressed		15	1239	1254
3	Grass synthetic		15	682	697
4	Trees		15	1229	1244
5	Soil		15	1227	1242
6	Water		15	310	325
7	Residential		15	1253	1268
8	Commercial		15	1229	1244
9	Road		15	1237	1252
10	Highway		15	1212	1227
11	Railway		15	1220	1235
12	Parking lot1		15	1218	1233
13	Parking lot2		15	454	469
14	Tennis court		15	413	428
15	Running track		15	645	660
Total			225	14804	15029

TABLE III
CATEGORY AND SAMPLE SETTINGS OF THE WHU-HONGHU DATASET

Class ID	Class name	Color	Training	Testing	Total
1	Red roof		30	14011	14041
2	Road		30	3482	3512
3	Bare soil		30	21791	21821
4	Cotton		30	163255	163285
5	Cotton firewood		30	6188	6218
6	Rape		30	44527	44557
7	Chinese cabbage		30	24073	24103
8	Pakchoi		30	4024	4054
9	Cabbage		30	10789	10819
10	Tuber mustard		30	12364	12394
11	Brassica parachinensis		30	10985	11015
12	Brassica chinensis		30	8924	8954
13	Small Brassica chinensis		30	22477	22507
14	Lactuca sativa		30	7326	7356
15	Celtuce		30	972	1002
16	Film covered lettuce		30	7232	7262
17	Romaine lettuce		30	2980	3010
18	Carrot		30	3187	3217
19	White radish		30	8682	8712
20	Garlic sprout		30	3456	3486
21	Broad bean		30	1298	1328
22	Tree		30	4010	4040
Total			660	386033	386693

was assessed using three standard evaluation metrics: overall accuracy (OA), average accuracy (AA), and the kappa coefficient (κ). To ensure statistical reliability, each experiment was independently repeated ten times with varying random seeds, and the results are presented as the mean accompanied by the standard deviation.

For all datasets, HGCNet adopts the same hyperparameter configuration unless otherwise specified. The hidden feature dimension, the state dimension of the selective scan module, and the expansion ratio are set to 64, 16, and 2.0, respectively. The network comprises three hierarchical stages for multilevel feature representation. The local spatial branch uses a 3×3 depthwise convolution, whereas the spatial attention gates in the interaction modules use 7×7 convolutions. These

TABLE IV
CATEGORY AND SAMPLE SETTINGS OF THE WHU-HANCHUAN DATASET

Class ID	Class name	Color	Training	Testing	Total
1	Strawberry		30	44705	44735
2	Cowpea		30	22723	22753
3	Soybean		30	10257	10287
4	Sorghum		30	5323	5353
5	Water spinach		30	1170	1200
6	Watermelon		30	4503	4533
7	Greens		30	5873	5903
8	Trees		30	17948	17978
9	Grass		30	9439	9469
10	Red roof		30	10486	10516
11	Gray roof		30	16881	16911
12	Plastic		30	3649	3679
13	Bare soil		30	9086	9116
14	Road		30	18530	18560
15	Bright object		30	1106	1136
16	Water		30	75371	75401
Total			480	257050	257530

hyperparameters are fixed across all datasets to ensure fair and reproducible comparisons.

C. Ablation Study

1) Network Blocks: We evaluate the contribution of the main blocks in HGCNet on four HSI datasets, and the results are summarized in Table V. The baseline is obtained by replacing each HSSI block with a plain 3×3 convolutional block without padding. The number of blocks is kept unchanged, and each block is followed by GN and GELU to ensure a fair comparison. This baseline gives the lowest accuracy on all datasets, showing that stacked local convolutions alone are not sufficient to learn discriminative spectral–spatial features. Adding HSRM consistently improves performance, which confirms the importance of stronger spatial context modeling. Using DSRM instead of HSRM also improves the baseline, indicating that explicit spectral recalibration contributes to classification. When HSRM and DSRM are used together, the gains become larger, suggesting that spatial and spectral modeling provide complementary information. The full model, which further includes BSSIM, achieves the best OA, AA, and κ on all four datasets. These results show that BSSIM improves the interaction and fusion of spectral and spatial cues, and that the three blocks contribute steadily to the final performance of HGCNet.

2) Internal Designs of HSRM and DSRM: To further investigate the internal designs of HSRM and DSRM, we evaluate the local-only and global-only variants of both modules, with the results reported in Table VI. For the analysis of HSRM, DSRM and BSSIM are kept unchanged, so the performance differences can be directly attributed to the design of HSRM. The same setting is used for DSRM. The results show that the global-only variant generally performs better than the local-only variant, indicating that global modeling plays a more dominant role in both spatial and spectral representation learning, while local modeling remains beneficial for preserving fine-grained structural and spectral details. Although the single-branch variants already achieve competitive performance, the full HGCNet still performs best, suggesting that neither local nor global modeling alone is sufficient and that more discriminative spectral–spatial representations are

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TABLE V

PERFORMANCE CONTRIBUTION OF DIFFERENT MODULES ON FOUR BENCHMARK DATASETS. BEST RESULTS ARE HIGHLIGHTED IN BOLD

HSRM	DSRM	BSSIM	Pavia			Houston			HongHu			HanChuan		
			OA	AA	κ	OA	AA	κ	OA	AA	κ	OA	AA	κ
✗	✗	✗	84.44	87.73	80.00	83.46	85.67	82.14	82.67	79.92	78.67	84.45	81.25	81.94
✓	✗	✗	87.34	90.19	82.21	89.11	88.03	89.08	90.47	89.36	88.09	88.09	86.84	86.17
✗	✓	✗	88.10	90.90	84.57	89.60	89.10	88.76	89.29	86.79	86.54	87.72	86.54	85.73
✓	✓	✗	90.70	90.21	89.95	89.94	89.44	88.85	91.11	88.86	88.80	89.70	86.69	87.97
✓	✓	✓	**93.13**	**91.05**	**94.71**	**90.61**	**89.85**	**92.14**	**92.33**	**90.29**	**92.10**	**92.35**	**91.08**	**91.29**

TABLE VI

ABLATION RESULTS OF THE INTERNAL DESIGNS OF HSRM AND DSRM ON FOUR HSI DATASETS IN TERMS OF OA (%). BEST RESULTS ARE HIGHLIGHTED IN BOLD

Module	Variant	Pavia	Houston	HongHu	HanChuan
HSRM	w/ Local	85.73	90.25	90.19	89.38
	w/ Global	87.19	90.14	91.31	91.37
DSRM	w/ Local	87.54	90.17	90.76	89.10
	w/ Global	88.00	90.52	90.14	90.76
HGCNet	Full Model	**93.13**	**90.61**	**92.33**	**92.35**

TABLE VII

ABLATION STUDY ON THE GATE MECHANISM AND INTERACTION DIRECTION IN BSSIM. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	Pavia	Houston	HongHu	HanChuan
w/o BSSIM	90.70	89.94	91.11	89.70
Spectral-to-spatial only	85.32	88.93	90.88	90.74
Spatial-to-spectral only	86.44	90.33	91.87	87.56
Bidirectional fusion w/o gate	91.93	89.95	91.46	89.85
Full BSSIM	**93.13**	**90.61**	**92.33**	**92.35**

obtained by jointly integrating local detail modeling and global context modeling.

3) Internal Design of BSSIM: To examine the internal design of BSSIM, we study the effects of the gate mechanism and the interaction direction, with the results reported in Table VII. Five settings are considered. In the "w/o BSSIM" setting, the BSSIM module is removed, and the spatial and spectral features are combined by element-wise addition. In the "Bidirectional fusion w/o gate" setting, the two branches are added and then passed through a 1×1 convolutional projection, while both the channel gate and the spatial gate are removed. This setting isolates the effect of adaptive gating in spectral–spatial fusion. For the two unidirectional settings, "Spectral-to-spatial only" keeps the guidance from the spectral branch to the spatial branch, where spectral features are used to generate channel attention and recalibrate spatial features. Conversely, "Spatial-to-spectral only" keeps the guidance from the spatial branch to the spectral branch, where spatial features are used to generate spatial attention and refine spectral features. The full BSSIM uses both directions, allowing spectral information to guide spatial representation learning and spatial structures to refine spectral representation learning.

As shown in Table VII, the full BSSIM achieves the best results on all four datasets. On Pavia, for example, the accuracy increases from 90.70% without BSSIM to 93.13% with the full BSSIM, showing the benefit of spectral–spatial interaction. The full BSSIM also outperforms bidirectional fusion without the gate mechanism on all datasets, indicating that simple bidirectional fusion followed by projection is not sufficient to model the complementarity between spatial and spectral representations. The gate mechanism enables adaptive feature selection and helps suppress less informative responses. In addition, the two unidirectional variants are inferior to the full BSSIM, suggesting that a single interaction direction cannot provide sufficient spectral–spatial guidance. Since spectral-to-spatial interaction mainly enhances spatial features according to spectral responses, while spatial-to-spectral interaction refines spectral features according to spatial structures, each direction captures only part of the complementary information between the two branches. These results support the use of both bidirectional interaction and adaptive gating in BSSIM.

4) Comparison With Channel Attention Mechanisms in DSRM: We further compare DSRM with two representative channel attention mechanisms, namely SE and CBAM, using the same backbone. As shown in Table VIII, adding channel attention improves the model without DSRM, suggesting that spectral channel recalibration is beneficial for HSI classification. However, the improvements from SE and CBAM remain limited. The full DSRM achieves the best OA on all four datasets. For example, it reaches 93.13% on the Pavia dataset, showing a clear gain over the competing variants. This result indicates that DSRM is more effective than conventional channel attention modules because it models both long- and short-range spectral dependencies, rather than relying only on global channel descriptors to recalibrate channel responses.

In terms of model complexity, DSRM introduces only a small number of additional parameters, increasing the parameter count from 544 to 560 K compared with SE and CBAM. Its FLOPs also remain close to those of the other variants. Although the inference time increases slightly, the accuracy gain is stable across the four datasets. These results show that DSRM offers a good balance between classification accuracy and computational cost.

D. Performance Comparison and Analysis

In this section, we compare the proposed HGCNet with several state-of-the-art (SOTA) HSI classification methods. A brief introduction to each method included in the experiments is summarized below.

1) *FcontNet [68]:* A multiscale classification method based on nonlocal contextual attention across spatial, spectral-band, and scale dimensions.
2) *SpectralFormer [28]:* A Transformer-based classification method that models the correlations among adjacent spectral bands and local spatial patches.

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TABLE VIII

ABLATION STUDY ON CHANNEL ATTENTION MECHANISMS IN DSRM. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Setting	Pavia	Houston	HongHu	HanChuan	Params (K)	FLOPs (G)	Inference Time (s)
w/o DSRM	85.95	89.14	89.86	89.13	**538**	**56.126**	**0.08**
SE block	88.19	89.53	90.28	90.16	544	56.143	0.08
CBAM block	88.65	89.27	90.84	90.43	544	56.152	0.09
Full DSRM	**93.13**	**90.61**	**92.33**	**92.35**	560	56.127	0.14

Note: The four dataset columns report OA (%). Params, FLOPs, and inference time are measured on the Pavia dataset under the same input size.

TABLE IX

QUANTITATIVE CLASSIFICATION RESULTS OF THE PAVIA UNIVERSITY DATASET. BEST RESULTS ARE SHOWN IN BOLD

Class	FcotNet	SpectralFormer	GAHT	MASSFormer	GSCViT	SpectralDiff	SS-Mamba	S2Mamba	MambaLG	MambaHSI	HyperSIGMA	HGCNet
1	76.32±4.15	61.12±7.68	90.05±3.83	93.69±1.77	93.80±2.43	82.40±3.33	**96.03±2.05**	87.25±3.52	93.18±4.87	87.02±1.95	79.99±2.81	92.34±2.36
2	83.12±4.10	67.94±7.16	87.42±5.40	87.48±8.38	84.77±7.04	87.01±2.25	85.95±5.47	78.74±8.57	83.65±5.15	**94.55±1.37**	92.79±4.54	91.51±4.64
3	89.78±5.27	76.57±8.87	86.02±5.19	90.77±4.21	95.33±3.02	92.65±2.55	**95.54±2.53**	82.57±5.88	90.84±4.26	89.32±3.70	94.92±3.44	91.20±4.41
4	64.19±13.71	89.92±3.04	95.64±0.76	95.79±1.45	95.02±1.41	72.38±4.17	**97.70±0.84**	95.48±0.84	96.00±0.92	79.36±1.35	77.75±2.86	89.25±2.29
5	97.37±3.09	99.89±0.27	99.60±0.52	99.62±0.51	99.28±0.93	99.54±0.36	**100**	99.62±0.44	99.92±0.25	99.67±0.20	97.61±1.81	**100**
6	92.46±6.09	69.59±8.53	93.31±4.76	94.38±5.17	96.23±3.45	94.92±1.26	94.17±7.47	88.35±8.33	96.39±3.99	98.52±0.87	**99.26±0.75**	99.21±1.16
7	92.49±2.28	74.74±7.27	98.18±1.20	98.43±1.79	99.51±0.55	92.07±1.89	**99.55±0.73**	96.27±1.95	98.40±1.46	87.73±0.90	97.68±2.70	96.66±0.95
8	82.90±12.48	63.81±9.42	93.62±3.86	91.65±4.01	**95.32±2.70**	88.65±1.93	93.26±4.44	91.69±4.56	92.25±3.90	90.51±3.36	91.62±3.15	93.61±3.34
9	73.23±6.62	95.60±2.23	95.34±2.48	98.26±0.91	97.29±1.05	84.96±6.25	**99.67±0.39**	94.45±2.27	98.95±0.46	93.31±0.45	74.29±5.81	98.61±1.05
OA (%)	82.63±3.09	70.53±3.78	90.46±2.66	91.32±3.96	90.86±3.49	87.10±1.10	91.58±2.35	85.22±3.78	89.90±2.46	92.08±0.70	90.39±1.93	**93.13±2.01**
κ (%)	77.68±3.88	63.00±4.27	87.67±3.29	88.81±4.84	88.27±4.30	83.22±1.40	89.13±2.92	81.19±4.43	87.03±3.06	89.57±0.90	87.45±2.39	**91.05±2.53**
AA (%)	83.54±2.74	77.69±1.65	93.24±1.11	94.45±1.58	95.17±1.39	88.29±1.25	**95.76±1.27**	90.49±1.02	94.40±1.17	91.11±0.55	89.54±0.95	94.71±0.93

3) *GAHT [48]:* A spectral–spatial classification method based on grouped pixel embedding and a hierarchical Transformer architecture.
4) *MASSFormer [46]:* A spectral–spatial classification method that incorporates prior knowledge through memory tokens and a memory-augmented encoder.
5) *GSCViT [51]:* A lightweight classification network utilizing groupwise separable convolution and ViT in each feature extraction block.
6) *SpectralDiff [32]:* A generative classification framework that combines spectral–spatial diffusion modeling with attention-based discriminative classification.
7) *SS-Mamba [57]:* A classification method based on spectral–spatial token modeling and stacked spectral–spatial Mamba modules.
8) *S2Mamba [62]:* A spectral–spatial classification method that achieves spectral–spatial scanning and feature fusion via a learnable hybrid gating mechanism.
9) *MambaLG [64]:* A spectral–spatial classification method based on dual-branch local–global modeling and dynamic spectral-aware fusion.
10) *MambaHSI [63]:* A classification method that models the entire image through a spectral–spatial dual-branch Mamba architecture and adaptively fuses spectral–spatial information.
11) *HyperSIGMA [34]:* A billion-scale vision transformer-based foundation model for unified HSI interpretation across diverse tasks and scenes.
12) *HGCNet:* The proposed HGCNet.

1) Quantitative Results: Tables IX–XII report the quantitative results on the Pavia University, Houston, WHU-HongHu, and WHU-HanChuan datasets, with the best results highlighted in bold. Overall, HGCNet obtains the highest OA on all four datasets, reaching 93.13%, 90.61%, 92.33%, and 92.35%, respectively. Compared with the strongest competing methods, HGCNet improves OA by 1.05%, 0.63%, 0.67%, and 1.04%, respectively, demonstrating its effectiveness and robustness under different scene complexities and category distributions. Among the CNN- and Transformer-based baselines, FcontNet introduces multiscale nonlocal contextual attention, but its attention-intensive and multiscale design may produce redundant representations. SpectralFormer models adjacent spectral bands and local spatial patches with a Transformer, yet its sequential band modeling remains limited in capturing rich long-range spectral–spatial dependencies. GAHT and MASSFormer improve contextual representation through hierarchical Transformer modeling and memory-augmented encoding, while GSCViT combines groupwise separable convolution with ViT to use both local and global information. Although these methods achieve competitive results, their performance remains constrained when handling large intraclass variability and heterogeneous spectral–spatial structures.

Recent advanced methods also show strong potential for HSI classification. SpectralDiff incorporates diffusion modeling into spectral–spatial representation learning and enhances classification by gradually refining discriminative features. However, its performance remains lower than that of HGCNet, such as 87.10% OA on Pavia University, indicating that diffusion-based representation refinement may still be insufficient for capturing complex spatial structures and subtle spectral variations. HyperSIGMA, as an FM-based method, benefits from large-scale hyperspectral pretraining and achieves strong performance, especially 91.31% OA on WHU-HanChuan. Nevertheless, it is still inferior to HGCNet on all four datasets, suggesting that task-specific hierarchical spectral–spatial modeling remains important for fine-grained HSI classification. Mamba-based methods further improve long-range dependency modeling with linear complexity. SS-Mamba, S2Mamba, MambaLG, and MambaHSI achieve competitive performance through different spatial–spectral scanning and fusion strategies, with MambaHSI and HyperSIGMA serving as the strongest competitors on several datasets. Nevertheless, these methods still lack sufficiently refined mechanisms for jointly preserving local structural details, modeling local band-wise variations, and enhancing

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TABLE X

QUANTITATIVE CLASSIFICATION RESULTS OF THE HOUSTON DATASET. BEST RESULTS ARE SHOWN IN BOLD

Class	FcotNet	SpectralFormer	GAHT	MASSFormer	GSCViT	SpectralDiff	SS-Mamba	S2Mamba	MambaLG	MambaHSI	HyperSIGMA	HGCNet
1	87.70±5.77	92.61±5.86	89.64±7.55	92.43±5.00	92.43±6.19	93.28±4.81	91.25±5.78	85.23±4.81	89.39±5.32	95.28±1.53	91.05±2.74	**97.78±0.73**
2	62.03±6.87	90.93±6.03	91.46±5.55	93.96±4.91	**94.73±4.66**	87.59±5.34	93.81±5.35	67.80±10.38	88.60±5.10	92.81±0.88	91.26±4.03	89.90±5.23
3	93.53±3.46	94.46±6.46	99.52±0.32	99.79±0.20	98.78±0.77	98.56±1.00	99.30±0.24	98.87±0.55	99.24±0.58	93.43±0.86	**99.80±0.19**	99.72±0.79
4	63.95±10.49	88.13±2.23	90.81±2.74	92.94±3.99	93.69±2.56	89.15±4.19	94.24±2.56	88.58±2.67	93.43±2.22	96.95±0.95	93.14±3.40	**97.92±0.85**
5	86.13±2.59	94.24±4.99	99.23±1.44	99.26±0.80	98.49±4.02	99.85±0.25	99.47±1.54	99.17±0.93	**99.97±0.05**	99.80±0.15	99.83±0.18	99.75±0.42
6	92.26±1.41	75.94±5.44	89.74±3.97	93.68±6.33	91.81±6.60	98.77±0.52	92.29±6.36	87.87±3.07	87.84±2.71	96.61±0.82	94.35±5.83	**99.10±1.00**
7	49.12±11.42	67.77±6.67	90.05±3.78	87.06±7.71	89.61±3.53	79.49±3.08	89.66±5.32	87.54±3.12	**91.25±4.93**	83.85±2.13	86.13±6.21	85.28±0.68
8	49.91±3.91	66.37±4.81	78.45±4.94	75.45±5.67	**79.21±5.42**	64.21±3.71	78.78±6.42	54.19±14.19	65.19±10.28	54.52±3.06	70.81±1.68	68.07±5.69
9	53.84±5.35	61.09±4.82	87.30±2.75	86.69±3.90	89.26±2.53	88.10±3.23	86.55±3.92	85.42±2.60	84.39±3.93	78.46±1.86	73.24±1.80	**90.08±1.68**
10	73.14±6.75	70.46±7.91	84.19±11.10	87.38±9.77	88.32±9.81	93.27±4.77	87.82±12.74	59.62±15.14	68.86±13.85	94.83±2.10	**96.27±2.18**	93.69±3.49
11	73.20±5.84	58.74±5.51	83.76±6.43	82.11±7.35	86.44±5.67	77.62±4.31	83.98±7.39	68.70±4.64	77.66±4.93	77.32±1.62	**93.44±3.70**	87.16±2.13
12	66.18±4.58	70.81±8.53	81.89±5.96	83.82±6.02	**83.54±7.33**	74.52±7.67	75.80±10.90	60.54±14.51	66.16±12.08	78.89±3.93	82.49±5.72	80.07±5.92
13	73.94±3.29	63.94±11.78	**96.70±2.20**	95.04±1.46	96.43±2.31	94.67±2.75	95.62±3.70	93.06±3.25	95.79±1.46	91.87±4.58	87.78±2.63	93.66±4.59
14	99.93±0.22	89.49±5.34	99.90±0.29	99.90±0.19	**100**	**100**	99.98±0.07	**100**	**100**	99.95±0.15	99.83±0.12	99.95±0.15
15	96.76±2.34	96.26±1.58	99.83±0.36	**100**	99.33±0.93	**100**	99.88±0.22	99.91±0.28	**100**	95.89±0.76	99.84±0.13	**100**
OA (%)	70.74±4.32	77.85±1.00	89.43±1.16	89.83±1.59	89.38±1.09	87.04±2.27	89.83±2.22	79.32±1.50	85.08±2.52	86.96±0.52	89.28±0.53	**90.61±0.93**
κ (%)	68.47±4.64	76.10±1.07	88.58±1.25	89.01±1.72	**90.25±1.18**	86.00±2.45	89.01±2.38	77.68±1.62	83.90±2.72	85.91±0.56	88.41±0.57	89.85±1.01
AA (%)	74.78±3.66	78.75±1.39	90.83±0.87	91.30±1.50	92.10±0.86	89.27±1.91	91.23±1.68	82.43±1.21	87.18±2.07	88.70±0.38	90.62±0.40	**92.14±0.74**

TABLE XI

QUANTITATIVE CLASSIFICATION RESULTS OF THE WHU-HONGHU DATASET. BEST RESULTS ARE SHOWN IN BOLD

Class	FcotNet	SpectralFormer	GAHT	MASSFormer	GSCViT	SpectralDiff	SS-Mamba	S2Mamba	MambaLG	MambaHSI	HyperSIGMA	HGCNet
1	91.17±2.43	93.32±1.74	95.45±0.87	**95.62±2.08**	94.51±2.37	85.92±1.32	92.34±7.34	93.75±3.33	94.11±3.60	93.47±2.12	90.67±4.39	90.43±3.05
2	88.76±1.62	87.02±3.13	95.50±2.23	94.58±2.43	94.70±2.15	88.08±2.12	**95.61±2.63**	94.24±4.10	92.15±3.63	92.12±1.68	78.87±4.87	94.92±1.42
3	86.11±0.42	78.32±3.17	83.60±6.07	86.07±4.08	85.07±5.27	86.34±1.06	84.78±7.82	79.08±2.53	82.15±6.87	**87.32±0.58**	84.47±2.08	85.12±0.63
4	89.41±2.43	95.93±2.36	96.28±1.85	95.58±2.86	94.38±2.98	89.73±2.58	95.72±1.87	96.63±1.45	95.61±2.28	97.30±1.41	97.03±0.94	**98.95±0.66**
5	93.55±1.68	90.01±5.39	93.84±3.10	91.62±7.91	90.96±3.63	89.66±1.71	93.20±9.19	94.28±1.76	90.01±6.44	93.62±1.23	**99.30±0.37**	88.27±4.15
6	90.75±0.72	91.73±2.88	92.75±2.21	**95.05±1.81**	92.51±2.83	89.98±0.76	92.92±5.85	90.14±3.48	91.57±2.96	91.93±1.23	92.22±1.73	92.58±0.76
7	72.68±2.22	73.08±2.63	81.53±4.07	**82.39±3.51**	80.51±4.91	65.82±1.68	81.20±4.87	76.80±4.93	79.04±5.60	77.93±1.40	78.05±1.24	73.69±1.91
8	97.79±0.81	72.06±3.49	80.19±6.16	84.52±4.44	77.35±4.90	82.77±3.46	77.45±11.01	74.92±5.84	73.57±3.30	96.58±1.26	**98.91±1.06**	88.43±1.50
9	91.01±1.20	95.59±1.22	**96.94±1.82**	96.58±2.08	96.22±1.79	89.01±1.78	96.59±1.88	96.33±1.83	95.15±1.84	91.32±0.73	91.00±2.14	95.37±0.78
10	83.29±2.07	70.16±4.70	**90.49±5.36**	88.74±5.79	85.91±5.09	79.64±2.90	87.59±10.17	85.19±3.85	81.50±8.71	86.35±2.51	79.79±0.60	90.33±0.62
11	69.15±3.54	75.64±8.42	86.67±5.17	**87.62±5.39**	81.62±5.61	61.42±3.69	86.26±5.92	75.49±5.92	79.88±7.09	83.32±2.06	79.14±6.50	72.52±1.16
12	86.23±2.56	70.05±3.89	81.58±5.58	83.98±5.82	78.97±4.01	81.45±1.40	82.56±7.45	76.50±3.89	75.08±2.95	**92.15±1.24**	80.57±1.41	91.42±0.82
13	83.01±2.22	72.97±3.24	78.90±2.13	78.98±3.54	72.65±3.08	79.16±2.06	75.51±6.63	79.07±1.51	77.24±3.62	**84.91±1.28**	72.50±4.62	67.25±6.07
14	**97.22±1.30**	91.29±1.90	96.58±1.40	96.44±2.10	94.55±1.88	78.29±1.22	94.10±5.24	95.32±1.54	92.11±2.26	89.32±3.44	79.48±8.05	80.10±5.04
15	96.22±0.88	96.71±1.93	99.66±0.44	99.05±0.78	98.87±0.56	98.21±0.86	**99.78±0.16**	99.23±1.04	99.17±0.48	97.07±1.32	94.90±1.28	97.09±1.15
16	88.13±2.70	93.33±3.04	95.02±2.66	96.59±2.68	96.90±1.42	82.63±2.42	94.49±6.94	89.35±5.02	92.81±8.66	95.57±0.92	89.64±0.49	**97.40±0.82**
17	96.01±0.34	91.17±3.66	98.19±1.52	98.63±1.69	96.29±1.61	98.55±1.00	98.85±1.21	79.47±18.61	95.48±3.63	**99.39±0.65**	95.41±1.46	99.25±0.69
18	**98.48±0.33**	92.70±3.24	96.97±2.14	97.66±1.58	97.59±2.14	89.87±2.96	97.85±1.68	96.53±1.84	98.15±1.40	97.33±0.83	98.45±2.00	96.09±0.48
19	87.48±1.77	89.77±3.18	93.44±2.69	93.33±4.05	91.98±1.92	85.60±0.82	**95.18±2.59**	91.66±2.28	91.80±3.08	94.03±0.55	83.24±4.20	95.12±0.27
20	96.52±1.19	93.53±2.16	94.89±3.08	**97.42±1.98**	96.81±1.42	84.68±4.78	94.68±9.99	96.81±1.55	96.58±1.31	96.60±0.90	96.44±3.60	93.03±0.52
21	99.98±0.07	94.03±2.92	94.31±8.39	98.45±1.58	97.03±5.67	99.65±0.46	89.17±29.31	97.13±2.20	90.26±12.91	99.96±0.05	**100.00±0.00**	98.72±1.01
22	96.04±1.09	96.08±2.73	96.79±3.12	97.61±3.26	95.18±3.88	96.14±2.33	97.48±2.02	97.95±2.51	96.60±3.47	96.88±1.46	**99.35±0.63**	96.24±1.27
OA (%)	87.76±1.31	88.80±1.36	91.14±1.26	91.40±1.44	90.43±1.15	85.59±0.96	91.50±2.32	90.42±0.70	90.31±1.31	91.66±0.71	90.45±0.43	**92.33±0.83**
κ (%)	84.91±1.55	85.99±1.62	90.12±1.53	90.51±1.72	88.07±1.36	82.21±1.09	89.38±2.87	87.99±0.84	87.89±1.59	**90.78±0.87**	88.01±0.52	90.29±1.03
AA (%)	89.95±0.76	86.57±0.76	90.80±0.82	**92.57±0.77**	90.48±0.44	85.57±0.45	91.06±3.99	88.90±0.89	89.09±0.87	91.48±0.43	89.07±0.79	92.10±0.72

bidirectional spectral–spatial interaction. In contrast, HGCNet adopts a hierarchical global–local collaborative framework, in which HSRM captures both long-range spatial dependencies and local structural details, DSRM models global spectral correlations and local interband variations, and BSSIM promotes bidirectional spectral–spatial guidance and complementary fusion. Benefiting from these designs, HGCNet consistently achieves the best OA across all datasets and obtains highly competitive κ, AA, and per-class accuracies, confirming its superiority and generalization capability for HSI classification.

2) Qualitative Results: As shown in Figs. 2–5, we compare the classification maps of representative methods on the Pavia University, Houston, WHU-HongHu, and WHU-HanChuan datasets. Most comparison methods produce generally reasonable classification maps, but they still exhibit varying degrees of visual inconsistency in complex regions. SpectralFormer and S2Mamba tend to generate scattered predictions, local confusion, and irregular boundaries, especially in complex urban scenes and agricultural areas with mixed land-cover classes. SpectralDiff alleviates part of the local noise through diffusion-based representation refinement, but scattered pixels and boundary confusion remain visible in regions with complex spatial structures. HyperSIGMA benefits from large-scale hyperspectral pretraining and provides competitive visual results, but local misclassification and incomplete region delineation can still be observed in several enlarged areas.

In contrast, HGCNet produces classification maps that are visually closer to the ground truth, with clearer boundaries, more homogeneous regions, and fewer isolated errors. These advantages are particularly evident on the Houston dataset, where roads, residential areas, and vegetation regions are highly mixed. Similar trends can also be observed on the Pavia University, WHU-HongHu, and WHU-HanChuan datasets, where HGCNet better preserves fine spatial structures and category transitions. These improvements are mainly attributed to the joint effect of HSRM, DSRM, and BSSIM, which enhance global–local spatial modeling, spectral dependency learning, and bidirectional spectral–spatial interaction, leading to more reliable classification maps in complex hyperspectral scenes.

3) Performance With Fewer Training Samples: We further investigate the performance of HGCNet under limited training samples by comparing it with several representative Mamba-based methods. As illustrated in Fig. 6, all methods benefit from more labeled samples, while HGCNet consistently yields the highest OA under all training-sample settings across the four datasets. This result suggests that HGCNet is less sensitive to training data scarcity and has stronger robustness in low-sample scenarios. The consistent advantage of HGCNet over the competing methods indi-

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TABLE XII

QUANTITATIVE CLASSIFICATION RESULTS OF THE WHU-HANCHUAN DATASET. BEST RESULTS ARE SHOWN IN BOLD

Class	FcotNet	SpectralFormer	GAHT	MASSFormer	GSCViT	SpectralDiff	SS-Mamba	S2Mamba	MambaLG	MambaHSI	HyperSIGMA	HGCNet
1	87.01±9.55	79.73±4.01	86.95±3.58	89.98±3.19	89.05±3.60	90.87±2.91	86.00±7.62	87.88±5.89	88.38±6.21	**96.04±0.58**	91.74±3.19	87.35±5.29
2	69.78±8.10	41.23±5.11	81.29±5.54	88.11±4.91	87.31±3.58	81.13±2.04	68.30±6.88	82.20±4.97	77.55±8.49	80.63±2.06	84.12±2.34	**88.27±2.65**
3	43.53±19.25	82.16±6.16	80.61±8.70	90.87±5.04	92.80±2.57	60.31±1.74	79.03±9.94	73.36±10.08	87.88±4.36	**97.69±0.36**	93.12±5.70	95.31±1.44
4	80.30±6.87	94.81±1.20	96.91±1.20	**98.92±0.85**	98.53±0.95	89.92±1.85	96.56±2.20	96.03±1.60	97.37±1.13	97.48±0.58	98.15±1.69	95.82±1.69
5	98.46±3.09	86.64±4.88	97.85±1.26	98.52±2.32	98.58±2.42	95.86±1.47	92.88±7.25	99.19±1.26	99.21±1.17	98.66±0.52	**99.74±0.21**	98.24±1.34
6	83.04±6.87	41.30±4.81	72.05±4.04	83.51±4.09	84.58±5.24	75.43±3.67	65.29±10.69	64.02±6.08	77.08±8.08	82.04±5.97	**94.84±2.56**	75.98±5.19
7	**97.09±2.60**	86.35±6.51	87.55±7.02	91.00±4.40	90.19±3.45	91.05±1.44	89.62±6.16	93.07±2.16	89.74±4.59	96.70±0.97	96.61±1.80	95.38±1.27
8	73.88±8.15	59.80±5.32	69.58±7.82	80.16±4.08	78.18±4.58	80.87±3.90	68.28±8.89	63.20±5.40	71.40±4.61	71.27±2.27	**86.02±2.82**	85.70±3.56
9	70.20±5.82	48.75±5.76	80.58±5.22	83.51±5.22	83.55±5.39	78.93±2.57	71.88±8.46	55.84±9.21	79.11±6.65	**92.33±1.55**	89.13±4.40	88.46±3.23
10	80.58±7.21	89.61±2.72	**97.39±1.65**	97.33±2.41	97.37±1.28	80.80±4.72	94.06±7.36	95.89±1.69	96.59±1.74	90.78±1.41	94.59±2.26	91.29±2.19
11	90.13±4.11	85.15±7.61	90.56±5.54	92.17±9.12	91.00±6.32	87.75±3.17	76.54±24.24	91.64±4.51	91.78±5.54	96.79±1.26	92.15±1.46	**97.07±1.01**
12	85.98±6.05	68.74±5.72	86.04±7.56	95.27±2.82	89.39±10.86	85.39±1.77	80.65±10.13	85.80±4.39	86.51±8.69	87.97±0.88	**98.27±1.23**	89.21±3.36
13	75.03±9.60	48.88±5.57	77.79±5.89	79.00±6.55	82.35±6.11	82.19±2.29	65.82±10.77	73.03±6.33	68.05±9.38	**89.36±0.41**	73.80±4.85	83.71±4.32
14	67.31±15.98	64.04±4.26	77.11±7.76	81.43±5.97	84.41±7.06	86.63±1.07	75.37±10.70	71.86±6.76	71.69±8.98	87.51±1.05	83.73±3.73	**91.46±2.36**
15	67.08±21.21	83.28±3.70	96.65±3.16	96.48±2.12	92.44±5.35	88.07±2.12	94.18±5.36	95.63±2.17	91.74±3.07	93.88±0.44	89.90±2.63	**98.35±0.49**
16	89.42±5.71	94.14±5.71	96.18±4.31	97.30±3.97	96.95±2.07	93.12±0.41	94.80±6.49	92.01±5.33	96.48±2.14	95.08±0.51	96.27±0.36	**99.04±0.62**
OA (%)	81.00±6.23	76.34±1.83	87.12±1.69	90.75±1.98	90.51±0.77	86.73±0.66	83.14±4.48	83.93±2.56	86.94±1.67	91.27±0.25	91.31±0.58	**92.35±0.93**
κ (%)	78.04±7.09	72.67±2.00	85.05±1.89	89.23±2.27	88.95±0.87	84.59±0.76	80.44±5.16	81.40±2.88	84.84±1.91	89.84±0.30	89.88±0.67	**91.08±1.08**
AA (%)	78.68±5.71	72.16±1.07	85.94±1.24	90.22±1.61	89.79±1.03	84.27±0.54	81.21±4.47	82.54±1.57	85.66±1.39	90.89±0.65	**91.39±0.36**	91.29±0.75

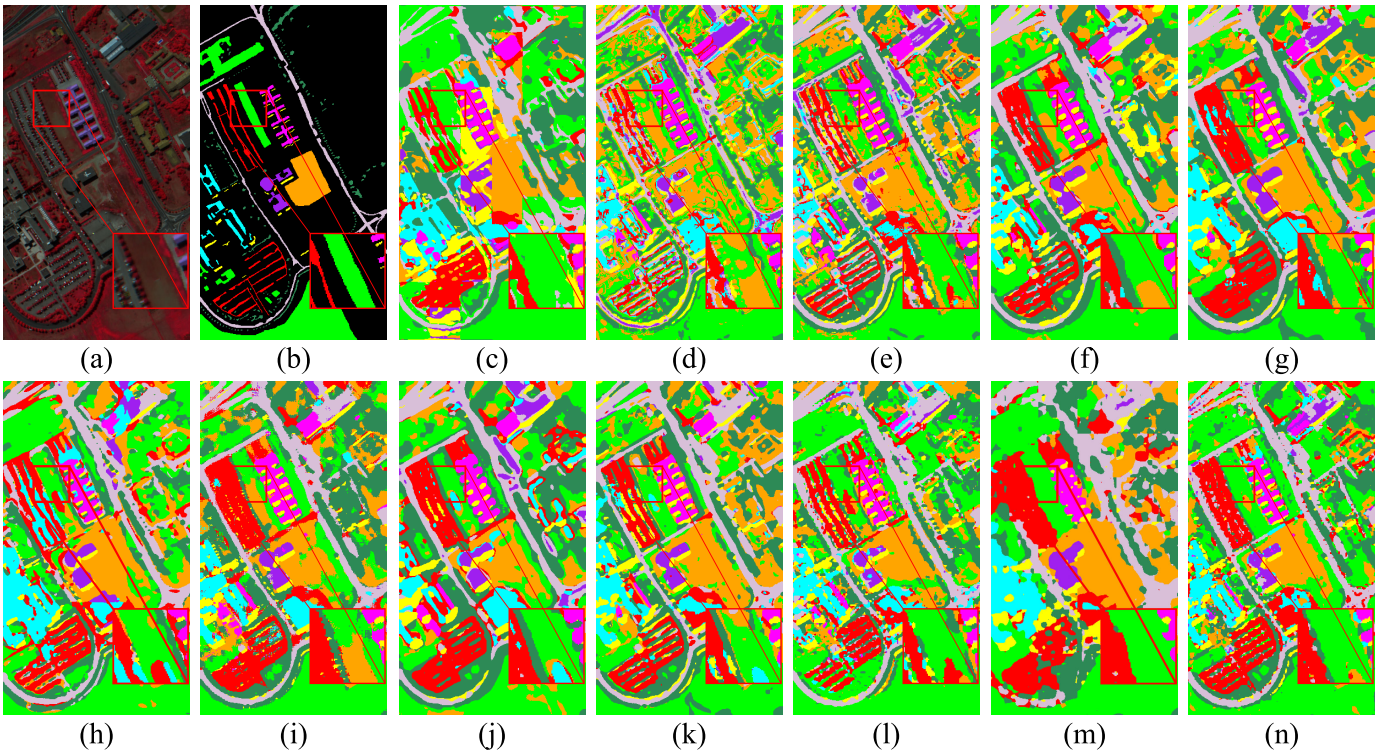


Fig. 2. Visualization of the classification maps on the Pavia University dataset. (a) False-color image. (b) Ground truth. (c) FcontNet. (d) SpectralFormer. (e) GAHT. (f) MASSFormer. (g) GSCViT. (h) SpectralDiff. (i) SS-Mamba. (j) S2Mamba. (k) MambaLG. (l) MambaHSI. (m) HyperSIGMA. (n) HGCNet.

TABLE XIII

COMPARISON OF MODEL COMPLEXITY, EFFICIENCY, AND PERFORMANCE ON THE PAVIA UNIVERSITY DATASET. MODEL PARAMETERS AND FLOPS ARE MEASURED WITH AN INPUT SIZE OF 103 × 13 × 13. THE BEST AND SECOND-BEST RESULTS ARE INDICATED IN BOLD AND UNDERLINED, RESPECTIVELY

Metrics	SpectralFormer	GAHT	MASSFormer	GSCViT	S2Mamba	MambaLG	MambaHSI	HGCNet
Parameters (M)	0.18	0.93	0.30	0.98	0.14	<u>0.13</u>	**0.03**	0.56
FLOPs (M)	<u>16.81</u>	156.62	51.02	131.27	24.79	30.96	**5.35**	45.75
Training time (s)	58.87	40.23	**19.29**	<u>25.00</u>	42.06	955.01	95.86	82.73
Inference time (s)	32.17	44.11	6.88	55.65	3.67	4.32	**0.13**	<u>0.14</u>
OA (%)	70.53	90.46	91.32	90.86	85.22	89.90	<u>92.08</u>	**93.13**

cates that the proposed model can more effectively exploit limited supervision to learn discriminative spectral–spatial representations. These results also verify that the integration of global contextual modeling and local detail modeling improves data efficiency and generalization ability in HSI classification.

4) Computation Cost Analysis: We further compare HGCNet with several Transformer- and Mamba-based methods in terms of model complexity, computational cost, training efficiency, inference efficiency, and classification performance, and the results are summarized in Table XIII. HGCNet achieves the highest OA of 93.13% on the Pavia Univer-

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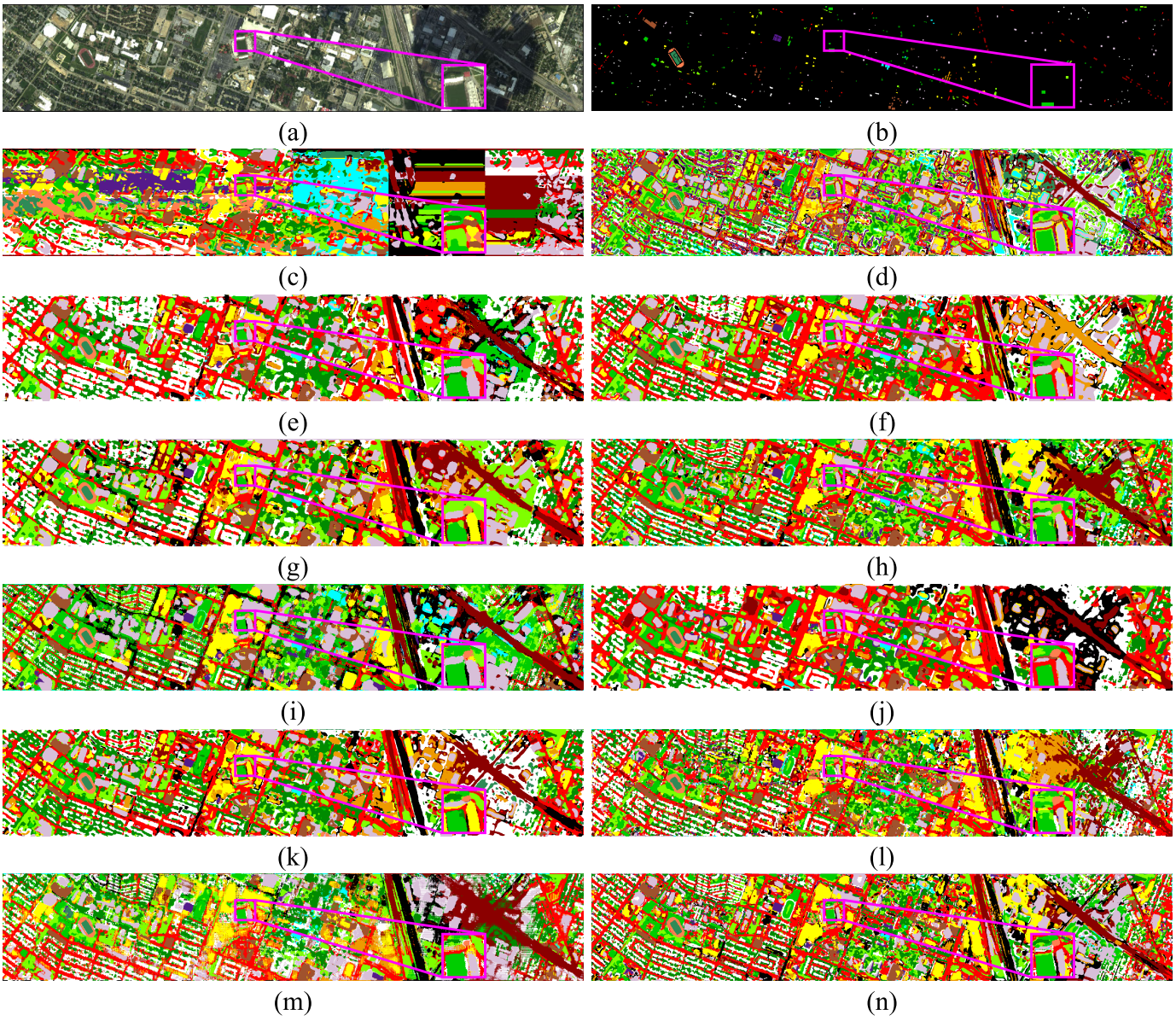


Fig. 3. Visualization of the classification maps on the Houston dataset. (a) False-color image. (b) Ground truth. (c) FcontNet. (d) SpectralFormer. (e) GAHT. (f) MASSFormer. (g) GSCViT. (h) SpectralDiff. (i) SS-Mamba. (j) S2Mamba. (k) MambaLG. (l) MambaHSI. (m) HyperSIGMA. (n) HGCNet.

sity dataset, outperforming all competing methods. Although HGCNet is not the smallest model in terms of parameter count or FLOPs, it still maintains a relatively compact size with 0.56-M parameters and a moderate computational cost of 45.75-M FLOPs. Both values are substantially lower than those of GAHT and GSCViT and remain within a practical range. In terms of efficiency, HGCNet attains an inference time of 0.14 s, ranking second and remaining very close to the best result of 0.13 s achieved by MambaHSI. This indicates that HGCNet supports fast full-image inference while preserving superior classification accuracy. Although its training time is not the shortest among all methods, it is markedly lower than that of MambaLG and remains competitive compared with several other baselines. These results show that HGCNet achieves a favorable balance between accuracy and efficiency. In particular, the hierarchical global–local collaborative design improves classification performance without introducing excessive computational burden, which demonstrates the practical effectiveness of the model.

E. Discussion on Advantages and Limitations

The experimental results show that the proposed HGCNet achieves competitive performance compared with existing HSI classification methods. Its main advantage lies in the joint consideration of global context, local details, and spectral–spatial interaction. Compared with CNN-based methods, HGCNet is better suited for modeling long-range dependencies in both spatial and spectral domains, while the convolutional branches in HSRM and DSRM help preserve local structures and subtle band-wise variations. Compared with Transformer-based methods, the use of Mamba-based sequence modeling reduces the computational burden associated with self-attention, making the model more efficient for spectral–spatial context

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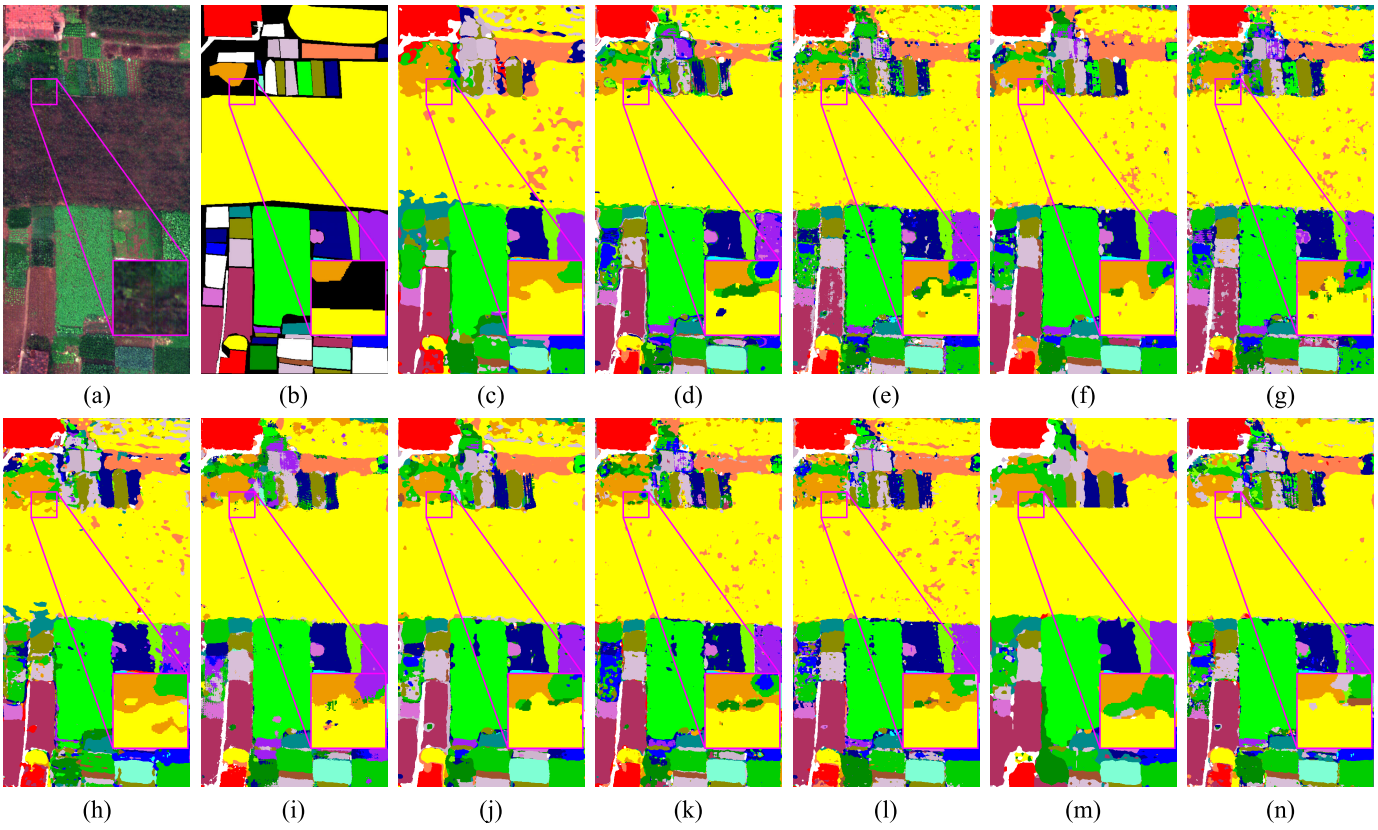


Fig. 4. Visualization of the classification maps on the Whu-HongHu dataset. (a) False-color image. (b) Ground truth. (c) FcontNet. (d) SpectralFormer. (e) GAHT. (f) MASSFormer. (g) GSCViT. (h) SpectralDiff. (i) SS-Mamba. (j) S2Mamba. (k) MambaLG. (l) MambaHSI. (m) HyperSIGMA. (n) HGCNet.

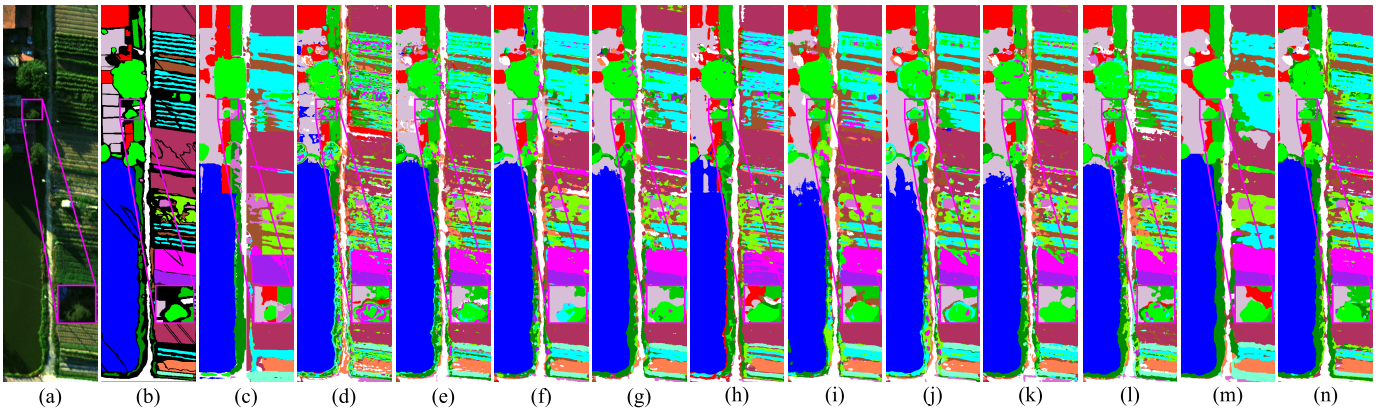


Fig. 5. Visualization of the classification maps on the Whu-HanChuan dataset. (a) False-color image. (b) Ground truth. (c) FcontNet. (d) SpectralFormer. (e) GAHT. (f) MASSFormer. (g) GSCViT. (h) SpectralDiff. (i) SS-Mamba. (j) S2Mamba. (k) MambaLG. (l) MambaHSI. (m) HyperSIGMA. (n) HGCNet.

modeling. In addition, compared with existing Mamba-based HSI classification methods that mainly focus on scanning strategy design or spectral–spatial joint modeling, HGCNet further introduces global–local collaborative modeling and bidirectional spectral–spatial interaction, which contributes to more discriminative feature representations.

Despite these advantages, the proposed method still has several limitations. First, the additional recalibration and interaction modules inevitably introduce extra parameters and computational cost compared with simpler CNN-based or single-branch Mamba models. Although the overall complexity remains manageable, further lightweight design is still worth exploring. Second, the current framework is mainly evaluated under standard supervised HSI classification settings. Its generalization ability under more challenging cross-scene, cross-sensor, or limited-label scenarios has not been fully investigated. Third, the performance of HGCNet may still depend on the quality and representativeness of the training samples, especially for categories with limited annotations or high intraclass variability. In future work, we will investigate more compact model structures and extend the proposed framework to domain adaptation and domain generalization settings for broader hyperspectral remote sensing applications.

V. Conclusion

This article proposed HGCNet, a method for HSI classification. Unlike existing methods that mainly emphasize

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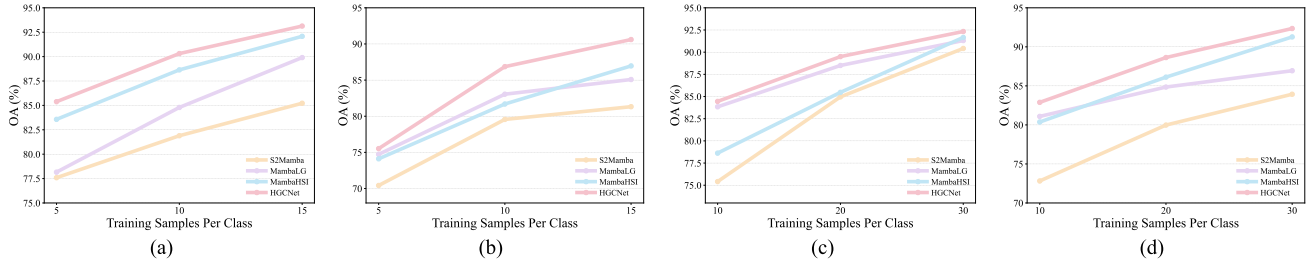


Fig. 6. Comparison of classification performance with fewer training samples. (a) Pavia University dataset. (b) Houston dataset. (c) Whu-HongHu dataset. (d) Whu-HanChuan dataset.

either global dependency modeling or local feature extraction, HGCNet explicitly coordinates both within a unified spectral–spatial learning framework, progressively optimizing spatial and spectral representations across hierarchical stages to characterize complex hyperspectral scenes more effectively. Within this framework, HSRM strengthens spatial representation learning by jointly modeling long-range contextual dependencies and local structural details, improving spatial semantics and fine-grained texture representation. DSRM enhances spectral feature learning through collaborative modeling of global spectral correlations and local band-wise variations, enabling more discriminative characterization of subtle interband differences. In addition, BSSIM promotes coordinated interaction between the spatial and spectral branches, allowing them to mutually refine each other and form more informative spectral–spatial representations. Extensive experiments on four benchmark datasets demonstrated that HGCNet consistently outperformed representative CNN-, Transformer-, and Mamba-based methods. Both quantitative and qualitative results verified its ability to produce accurate, robust, and spatially coherent classification maps, indicating that jointly exploiting global contextual information, local structural details, and bidirectional spectral–spatial interaction is an effective paradigm for HSI classification.

Although HGCNet achieves promising results, its additional recalibration and interaction modules still introduce extra computational overhead, and its generalization ability under more complex cross-scene or cross-sensor conditions remains to be further examined. Future work will focus on developing more lightweight variants of HGCNet and extending the framework to domain adaptation, domain generalization, and limited-label hyperspectral classification scenarios.

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